

Practical 7(Lagrange Interpolation)

P-1:

```
In[*]:= LagrangePolynomial[x0_, f0_] :=  
Module[{xi = x0, fi = f0, n, m, polynomial},  
  n = Length[xi];  
  m = Length[fi];  
  If[n ≠ m, Print["List of points and functions values are not of same size"];  
  Return[]];  
For[i = 1, i ≤ n, i++,  
  L[i, x_] = (Product[(x - xi[[j]]) / (xi[[i]] - xi[[j]]), {j, 1, i - 1}]) *  
    (Product[(x - xi[[j]]) / (xi[[i]] - xi[[j]]), {j, i + 1, n}]);  
polynomial[x_] = Sum[L[k, x] * fi[[k]], {k, 1, n}];  
Return[polynomial[x]]];
```

Question 1

```
In[*]:= nodes = {0, 1, 3};  
values = {1, 3, 55};  
lagrangePolynomial[x_] = LagrangePolynomial[nodes, values]
```

```
Out[*]=
```

$$\frac{1}{3} (1 - x) (3 - x) + \frac{3}{2} (3 - x) x + \frac{55}{6} (-1 + x) x$$

```
In[*]:= Expand[%]
```

```
Out[*]=
```

$$1 - 6x + 8x^2$$

Question 2:

```
In[*]:= nodes = {0, 1, 3};  
values = {1, 3};  
lagrangePolynomial[x_] = LagrangePolynomial[nodes, values]
```

List of points and functions values are not of same size

Question3:

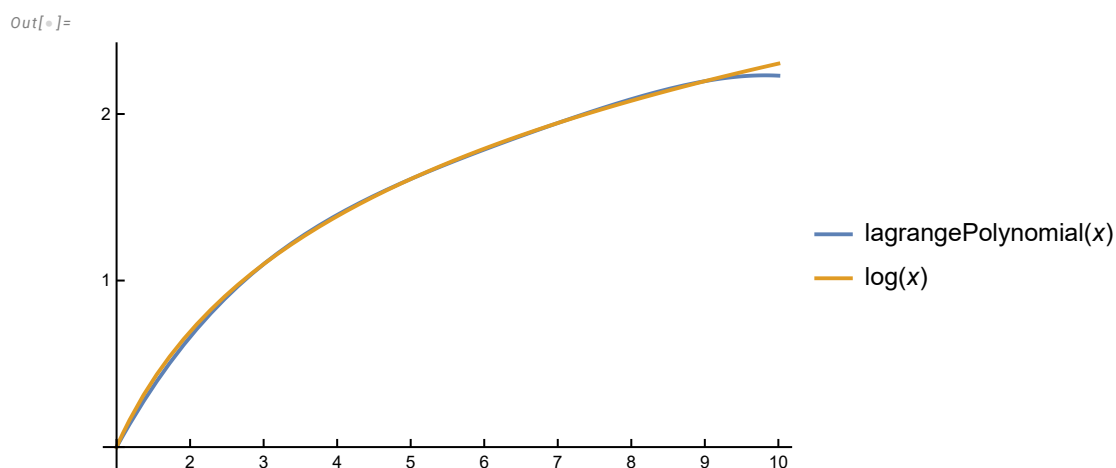
```
In[*]:= nodes = {1, 3, 5, 7, 9};
values = {N[Log[1]], N[Log[3]], N[Log[5]], N[Log[7]], N[Log[9]]};
lagrangePolynomial[x_] = LagrangePolynomial[nodes, values]
```

```
Out[*]=
0. + 0.0114439 (5 - x) (7 - x) (9 - x) (-1 + x) + 0.0251475 (7 - x) (9 - x) (-3 + x) (-1 + x) +
0.0202699 (9 - x) (-5 + x) (-3 + x) (-1 + x) + 0.00572194 (-7 + x) (-5 + x) (-3 + x) (-1 + x)
```

```
In[*]:= Simplify[%]
```

```
Out[*]=
-0.987583 + 1.18991 x - 0.223608 x^2 + 0.0221231 x^3 - 0.000844369 x^4
```

```
In[*]:= Plot[{lagrangePolynomial[x], Log[x]}, {x, 1, 10},
Ticks -> {Range[0, 10]}, PlotLegends -> "Expressions"]
```



Question 4:

```
In[*]:= nodes = {-1, 0, 1, 2};
values = {5, 1, 1, 11};
lagrangePolynomial[x_] = LagrangePolynomial[nodes, values]
```

```
Out[*]=
-5/6 (1 - x) (2 - x) x + 1/2 (1 - x) (2 - x) (1 + x) + 1/2 (2 - x) x (1 + x) + 11/6 (-1 + x) x (1 + x)
```

```
In[*]:= Simplify[%]
```

```
Out[*]=
1 - 3 x + 2 x^2 + x^3
```

```
In[*]:= lagrangePolynomial[1.5]
```

```
Out[*]=
4.375
```

